Food Tests

Benedicts test for reducing sugars

Benedicts Test for Reducing Sugars
Procedure for liquid: 1. add an equal volume of Benedicts solution to the sample in a test tube 2. Shake thoroughly 3. Place the test tube in a beaker of boiling water for 2-3 minutes
Procedure for solid: 1. add finely chopped sample into a test tube containing 2cm3 of denoised water 2. Shake thoroughly 3. Allow solids to settle 4. decant the liquid into another test tube 5. Add equal volume of benedicts solution into the test tube 6. Shake thoroughly 7. Place test tube in a beaker of boiling water for 2-3 minutes
Positive result: benedicts solution turns from blue to green/orange/yellow/red precipitate upon heating
Negative result: Benedicts solution remains blue upon heating

Solution remained blue	No reducing sugars
Solution turned from blue to green precipitate	traces of reducing sugars
solution turned from blue to yellow/orange precipitate	moderate amount of reducing sugars
solution turns from blue to red precipitate	large amounts of reducing sugars

Reducing Sugars	Non-Reducing Sugars
- glucose - fructose - Maltose - Lactose - Glyceraldehyde - Arabinose	- Stachyose - Sucrose - Tehalose - Verbascose - Raffinose

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Test for Starch

Add a few drops of iodine solution to sample on a white tile

positive result

iodine solution turns from yellow/brown/yellowish-brown to blue back

negative result

iodine solution remains yellow/brown/yellowish-brown

Alcohol Emulsion Test for Fats

Alcohol Emulsion Test for Fats
Procedure for Liquid 1. add 2Cm3 of ethanol to a drop of the sample 2. Shake thoroughly 3. Add the solution to 2cm3 of water in a test tube
Procedure for Solid 1. add finely chopped sample into a test tube containing 2cm3 of ethanol 2. shake thoroughly 3. Allow solid to settle 4. Decant the liquid into another test tube containing 2cm3 of water.
Positive Result: A cloudy white emulsion is formed (remember to state the change)
Negative Result: Solution remains clear

Biurets Test for Proteins

Biurets Test for Proteins
Procedure for Liquid: 1. add an equal volume of sodium hydroxide to the sample in a test tube 2. add a drop of Copper(II)sulfate solution - shake after each drop OR 1. add an equal volume of biurets solution to sample in a test tube 2. shake well and allow mixture to stand for 5 minutes
Procedure for Solid 1. Add finely chopped sample into a test tube containing 2cm3 of deionised water 2. shake thoroughly 3. Allow solids to settle 4. Decant the liquid into another test tube

Biurets Test for Proteins
5. Add an equal volume of sodium hydroxide to sample in the test tube 6. add a drop of Copper(II)sulfate solution - shake after each drop
Positive Result: Solution turns from blue to violet
Negative result: Solution remains blue